

OPTICAL-FIBRE SENSORS Shaping The New Age Of Connectivity

In this era of communication and connectivity, optical-fibre sensors find many applications, some of which are discussed in this article



Vinayak Ramachandra Adkoli is BE in industrial production. He was lecturer in mechanical department for 10 years in three different polytechnics. Now, he is a freelance writer and cartoonist

For ages, sensors have been used in hazardous environments. These sensors often contain electronic components. In the past, it was a challenge for engineers to make sensors work at extremely high temperatures, such as in industrial furnaces and sintering operations. A lot of extra effort went into active and passive sensors, to keep temperature in check, as well as complex verification, characterisation and qualification of sensors—result being the innovation of optical-fibre (or fibre-optic) sensors.

Fundamentally, fibre-optic sensors make use of optical fibre as an essential element to transfer and amplify the quantity to be measured. Optical fibre is an important element in modern electronics communication such as broadband Internet. Even today, optical fibre can be used in sensor technology to increase accuracy and capability.

Why integrate optical fibre in sensor technology

Optical fibre makes use of optical principle, and is based on total internal reflection of light. It was developed in 1970 by Corning Glass Works. At the same time, GaAs semiconductor lasers were also developed for transmitting light through fibre-optical cables.

An optical-fibre cable consists of core and cladding, both of which have different refractive indices. Core has a high refractive index, while cladding has a low refractive index.

An optical-fibre cable receives light beam as input, which is nothing but the quantity to be measured, as sensed by input

of the sensor. Input for the cable is provided by a broadband light source such as LED, or narrow-band light source such as laser diode. This light beam travels through the optical fibre without any loss. It is dispersed at an angle of 60 degrees and emitted to the target (output of the sensor).

Core may be of two types: plastic or glass.

Types of optical-fibre sensors

Optical-fibre sensors, or fibre-optic sensors, are used to measure quantities such as temperature, pressure, vibration, displacement, rotation or concentration of chemicals. These are available in tiny sizes, and can be used in remote sensing, as these require no electric power.

Optical-fibre sensors can be classified as follows:

- According to location of sensor: intrinsic fibre-optical and extrinsic fibre-optical
- Based on operating principles: intensity, phase and polarisation
- Based on application: chemical, physical and bio-medical

Intrinsic type. In this type of sensor sensing takes place within the fibre. This means that the fibre is directly affected by the measured quantity.

Extrinsic type. Here, the optical fibre is used as the information carrier. These sensors are used to measure rotation, vibration, velocity, displacement, twisting, torque and acceleration. The best example of this type of sensor is the temperature sensor inside an aircraft jet engine. It uses optical fibre to transmit radiation into a radiation pyrom-